Probability and Statistics course

Introduction

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ELTE

March 19, 2026

Week 6

Midterm

- There will be a midterm worth of 50 points on October 20 and a exam worth 50 points on December 8 during the class. . Thresholds for the grades:
  - 0 – 34 : 1
  - 35 – 52 : 2
  - 53 – 65: 3
  - 66 – 78: 4
  - 79 – 100: 5.
- Both exams and mid-term are mandatory. To pass them, you need to obtain at least 30% from both of them. If you do not manage, you need to participate in the retake. There is only one possibility for a retake. The only cases where you are allowed not to participate in the midterm/exam is when you are ill, and then I expect you to bring me a proper document from the doctor confirming that you were not able to appear at the faculty due to illness, or when you have another exam at the same time, and then you need to bring a confirmation from another professor.
- If you bring me such a document, I will organize 2 retakes for you if needed (so that everyone has 2 attempts).
- In case you cannot appear on the midterm/exam and you fail to bring the proper documentation justifying your absence, I will organize the so-called "utovizsga" at the end of the semester for you, which is a big exam consisting of the material from the whole semester.

Covariance

Definition (Covariance)

Covariance of two random variables:

$$\operatorname{cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

The joint density function of (X, Y) is $f(x, y)$ if for every appropriate subset $A \subseteq \mathbb{R}^2$ we have $\mathbb{P}(X \in A) = \int_A f(x, y)\, dx\, dy$. Calculation of covariance if the joint density function is $f(x, y)$:

$$\operatorname{cov}(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy \cdot f(x, y)\, dx\, dy - \left(\int_{-\infty}^{\infty} x \cdot f_X(x)\, dx \right) \cdot \left(\int_{-\infty}^{\infty} y \cdot f_Y(y)\, dy \right),$$

where $f_X(x)$ and $f_Y(y)$ are the marginal densities that can be computed as $f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$ and $f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$.

Correlation

Definition (Pearson Correlation)

The (Pearson) correlation coefficient of X and Y (this is always at least -1 and at most 1):

$$R(X,Y) = \frac{\mathrm{cov}(X,Y)}{sd(X)sd(Y)}.$$

$R(X,Y) = 1$ means that $Y = aX + b$ for some a, b, with $a > 0$

Examples

A shop is selling smaller and larger cups of coffee. The price of the small version is 200, the price of the large version is 300. Let X be the number of small coffees sold in a day, and Y the number of large coffees sold in a day. We assume that X and Y are independent, have Poisson distribution, and both have expectation 80. Determine the correlation coefficient of the total number of coffees sold in a day and the income (the total price of coffees sent in a day). What will be different if the price of the large version is not 300, but $350, 400, 450$ etc.?

Solution

The highest possible value of the correlation coefficient is 1, hence this means a strong positive correlation (the larger the total number is, the larger the income is), that is, a strong linear dependence. Correlation coefficients measures linear dependence. First let us show some useful general properties:
$\text{cov}(U, 200V) = \mathbb{E}(U \cdot 200V) - \mathbb{E}(U)\mathbb{E}(200V) = 200\text{cov}(U, V)$
If $U = V$: $\text{cov}(U, U) = \mathbb{E}(U^2) - \mathbb{E}(U)^2 = Var(U)$, this is the variance. First we calculate the covariance. We can use that covariance is linear in both variables, and that the covariance of independent random variables is 0:

$$\begin{aligned}\text{cov}(X + Y, 200X + 300Y) &= \text{cov}(X, 200X) + \text{cov}(Y, 200X) + \text{cov}(X, 300Y) + \text{cov}(Y, 300Y) = 200\text{cov}(X, X) + 200\text{cov}(X, Y) + 300\text{cov}(X, Y) + 300\text{cov}(Y, \\ &= 200Var(X) + 300Var(Y) = 500 \cdot 80 = 40000,\end{aligned}$$

because the variance of the Poisson distribution with parameter λ is equal to λ, and X, Y have parameter λ and $(a + b)(c + d) = ac + ad + bc + bd$.
Or alternatively, we may use the definition and the properties of the expectation: $\mathbb{E}(200X + 300Y) = 200\mathbb{E}(X) + 300\mathbb{E}(Y)$

$$\mathbb{E}((X + Y)(200X + 300Y)) = \mathbb{E}(200X^2 + 500XY + 300Y^2) = \mathbb{E}(200X^2) + \mathbb{E}(500XY) + \mathbb{E}(300Y^2)$$

$$\begin{aligned}\text{cov}(X + Y, 200X + 300Y) &= 200\mathbb{E}(X^2) - 200\mathbb{E}(X)\mathbb{E}(X) + 500\mathbb{E}(XY) - (200 + 300)(\mathbb{E}(X) \cdot \mathbb{E}(Y)) \\ &+ 300(\mathbb{E}(Y^2) - \mathbb{E}(Y)^2) = 200Var(X) + 500\text{cov}(X, Y) + 300Var(Y) = 40000\end{aligned}$$

$$\begin{aligned}R(X + Y, 200X + 300Y) &= \frac{\text{cov}(X + Y, 200X + 300Y)}{sd(X + Y)sd(200X + 300Y)} == \frac{200Var(X) + 300Var(Y)}{\sqrt{Var(X) + Var(Y)}\sqrt{200^2 Var(X) + 300^2 Var(Y)}} = \frac{200Var(X) + 300Var(Y)}{\sqrt{2}sd(X) \cdot \sqrt{200^2 + 300^2}sd(X)} = \\ &= \frac{500}{\sqrt{2} \cdot \sqrt{130000}} = 0.9806,\end{aligned}$$

because if X, Y are independent, then $sd(aX + bY) = \sqrt{a^2 Var(X) + b^2 Var(Y)}$,and now we have $Var(X) = Var(Y) = 80$. Similarly, if the price of the large cup is c:
$R(X + Y, 200X + cY) = \frac{\text{cov}(X+Y,200X+cY)}{sd(X+Y)sd(200X+cY)} = \frac{200Var(X)+cVar(Y)}{\sqrt{2}sd(X)\cdot\sqrt{200^2+c^2}sd(X)} = \frac{200+c}{\sqrt{2}\cdot\sqrt{40000+c^2}}$.
Correlation coefficient measures the strength of linear dependence. If it is close to 1, then the points are close to a line with positive slope. If it is close to -1, then the points are close to a line with negative slope.

Corr and Cov examples

Examples

Suppose that the joint density function of (X, Y) is given as follows.

$$f(x, y) = \begin{cases} x + y, & \text{if } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1; \\ 0 & \text{otherwise.} \end{cases}$$

Determine the correlation coefficient of X and Y.

Solutions

We first determine the expectation of the product.

$$\mathbb{E}(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy \cdot f(x,y)\, dy\, dx = \int_0^1\int_0^1 xy(x+y)dydx = \int_0^1\int_0^1 x^2ydydx + \int_0^1\int_0^1 xy^2dydx = \int_0^1 \frac{x^2}{2}dx + \int_0^1 \frac{x}{3}dx = \frac{1}{6}+\frac{1}{6} = \frac{1}{3},$$

by using that $\int_0^1 x^k dx = [x^{k+1}/(k+1)]_{x=0}^1 = 1/(k+1)$. The density function of X is as follows:
$f_1(x) = \int_{-\infty}^{\infty} f(x,y)dy = \int_0^1 x+y\, dy = x + \left[\frac{y^2}{2}\right]_{y=0}^1 = x + \frac{1}{2}$, if $0 \leq x \leq 1$. Otherwise we integrate 0, and the integral is also zero. This corresponds to the fact that all possible values of (X, Y) are in the square $[0,1] \times [0,1]$.
From the density function we have:

$$\mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 f_1(x)dx = \int_0^1 x^2 \cdot \left(x + \frac{1}{2}\right)dx = \int_0^1 x^3 + \frac{x^2}{2}dx = \frac{1}{4}+\frac{1}{6} = \frac{5}{12}.$$

By symmetry, for Y we get the same expectations. We obtain that

$$\mathbb{E}(XY) = \frac{1}{3}; \qquad \mathbb{E}(X) = \mathbb{E}(Y) = \frac{7}{12}; \qquad \mathbb{E}(X^2) = \mathbb{E}(Y^2) = \frac{5}{12}.$$

Finally, the correlation coefficient of X and Y is equal to:

$$R(X,Y) = \frac{\mathrm{cov}(X,Y)}{sd(X)sd(Y)} = \frac{\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)}{sd(X)sd(Y)} = \frac{1/3 - (7/12)^2}{\sqrt{5/12 - (7/12)^2} \cdot \sqrt{5/12 - (7/12)^2}} = \frac{1/3 - (7/12)^2}{5/12 - (7/12)^2} = -0,091.$$

This means that there is no significant correlation between the two quantities.

Examples

1. There are 15 boys and 18 girls in a school class. Suppose that students are absent from school independently of each other, everyone with probability $1/10$ on a given day. Determine the covariance and correlation coefficient of the number of girls who are absent and the total number of absent students.
2. We measured temperature at a given place with two different tools. Suppose that the results are independent, normal variables with expectation 6 and standard deviation 2. Let X be the first result, Y the second (with the second tool). Determine the following quantities: $\mathrm{cov}(X, X+Y), \mathrm{cov}(X, \frac{X+Y}{2})$, $\mathrm{cov}(X-Y, X+Y)$, $R(X, \frac{X+Y}{2})$.
3. Suppose that the joint density function of (X, Y) is given as follows.

$$f(x,y) = \begin{cases} x+y, & \text{if } 0 \le x \le 1 \text{ and } 0 \le y \le 1; \\ 0 & \text{otherwise.} \end{cases}$$

   Determine the correlation coefficient of X and Y.

Solutions

1. X: number of girls who are absent; Y: number of boys who are absent. Here X has binomial distribution with parameters $n = 18$ and $p = 0.1$, while Y has binomial distribution with parameters $n = 15$ and $p = 0.1$. In addition, X and Y are independent of each other, hence their covariance is 0.

$$\mathrm{cov}(X, X+Y) = \mathrm{cov}(X, X) + \mathrm{cov}(X, Y) = Var(X) = 18 \cdot 0.1 \cdot 0.9 = 1.62.$$

Recall that variance is equal to $np(1-p)$ for binomail distribution.

$$R(X, X+Y) = \frac{\mathrm{cov}(X, X+Y)}{sd(X)sd(X+Y)} = \frac{Var(X)}{sd(X)sd(X+Y)} = \frac{sd(X)}{sd(X+Y)} = \frac{\sqrt{18 \cdot 0.1 \cdot 0,9}}{\sqrt{33 \cdot 0.1 \cdot 0.9}} = \frac{\sqrt{18}}{\sqrt{33}} = 0.7385.$$

This corresponds to a positive correlation of medium strength (1 is the largest possible value).

2.

$$\mathrm{cov}(X, X+Y) = Var(X) + 0 = 4.$$

$$\mathrm{cov}(X, \frac{X+Y}{2}) = \frac{1}{2}(Var(X) + 0) = 2.$$

$$\mathrm{cov}(X-Y, X+Y) = Var(X) - Var(Y) = 0.$$

$$R(X, \frac{X+Y}{2}) = \frac{\mathrm{cov}(X, \frac{X+Y}{2})}{sd(X)sd(\frac{X+Y}{2})} = \frac{\frac{1}{2}Var(X)}{\frac{1}{\sqrt{2}}Var(X)} = \frac{1}{\sqrt{2}} = 0.7071$$

Solutions

3 We first determine the expectation of the product.

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Finally, the correlation coefficient of X and Y is equal to:

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Example Midterm

1. For which $n > 1$ will be independent
   a) the following events: A: there is at least one head and at least one tail, B: there is at most one tail from n tosses of a coin.
   b) the following events: A: there is at least one head and at least one tail and B: the first result is head, from n flips of a coin.
2. A medical examination result shows condition "A" (e.g. high cholesterol level) with probability 0.05 and condition "B" (e.g. low iron level) with probability 0.03. The probability of joint occurrence is 0.01.
   a) Are the two conditions independent? If not, what would be the probability of joint occurrence in case of independence?
   b) What is the probability that a randomly chosen person has neither of the two conditions?
   c) What is the probability of finding condition "B" among those people with condition "A"? Conversely, what is the probability of finding condition "A" among those people with condition "B"?
3. Let us assume that the number of faults against a given player during a basketball game has Poisson distribution with parameter $\lambda = 2$ and that each of these faults is noticed by the referee with probability 0.5 (independently from the others). Compute the distribution of the noticed faults!
4. Let the density function of X $f(x) = 1 - |x|$ for $-1 < x < 1$ (and 0 otherwise). Compute $E(X)$ and $Var(X)$!
5. Let us assume that the strength of a rope is a normally distributed random variable with expectation 1000 kN. What can the standard deviation be if we know that the probability of a rope to be weaker than 980 kN is 2%?
6. Let X and Y be independent random variables with mean 0 and variance 1 Give a value c such that $(3X - cY, 2X + Y) = 0$.

Solutions

1. a $P(A) = P(\text{we have flipped both heads and tails}) = 1 - P(\text{we have only one result}) = 1 - 2P(\text{we have only heads}) = 1 - 2\frac{1}{2^n} = 1 - \frac{1}{2^{n-1}}$

   $P(B) = P(\text{we have flipped at most one tail}) = P(\text{we have 0 tails}) + P(\text{we have flipped exactly one tail}) = \frac{1}{2^n} + \binom{n}{1}\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^{n-1} = \frac{n+1}{2^n}$

   $P(A \cap B) = P(\text{we have flipped exactly one tail}) = \frac{n}{2^n}$

   The following equality has to be solved in n: $P(A \cap B) = P(A)P(B)$, which results in $n + 1 = 2^{n-1}$. It is easy to see that it is true only for $n = 3$.

   b $P(A) = 1 - \frac{1}{2^{n-1}}$

   $P(B) = P(\text{the first flip is head}) = \frac{1}{2}$

   $P(A \cap B) = P(\text{the first flip is head, and there is a tail in the rest}) \overset{\text{independence}}{=} P(\text{the first is head})P(\text{there is a tail in the rest})$
   $\frac{1}{2}(1 - P((n-1)\text{ head})) = \frac{1}{2}\left(1 - \frac{1}{2^{n-1}}\right)$

   The equality $P(A \cap B) = P(A)P(B)$ has to be solved. It leads to $\frac{1}{2}\left(1 - \frac{1}{2^{n-1}}\right) = \left(1 - \frac{1}{2^{n-1}}\right)\frac{1}{2}$ which holds for every $n > 1$, so they are independent.

2. Let A = the person has condition A; B = the person has condition B

   a) If A and B were independent events, then $P(A \cap B) = P(A)P(B) = 005 \cdot 0.03 = 0.0015$ should hold instead of 0.01, so A and B are not independent.

   b) The complement event is that at least one of the conditions is observed. Thus
   $P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - (P(A) + P(B) - P(A \cap B)) = 1 - (0.05 + 0.03 - 0.01) = 0.93.$

   c) $P(A|B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.01}{0.03} \approx 0.33$, and on the other way around: $P(B|A) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{0.01}{0.05} = 0.2.$

Solutions

3 Let X denote the faults and Y the observed ones.

$$P(Y=k)=\sum_{m=k}^{\infty}P(Y=k|X=m)P(X=m)=\sum_{m=k}^{\infty}\binom{m}{k}0.5^m\lambda^m\frac{e^{-\lambda}}{m!}=\left(\frac{\lambda}{2}\right)^k\frac{e^{-\lambda}}{k!}\sum_{m-k=0}^{\infty}\left(\frac{\lambda}{2}\right)^{m-k}\frac{1}{(m-k)!}$$

and the sum gives $e^{\lambda/2}$, so the result is $\left(\frac{\lambda}{2}\right)^k\frac{e^{-\lambda/2}}{k!}$, which is Poisson distribution with parameter $\lambda/2$.

4 $E(X)=0$, as we have a symmetric density with bounded support. $Var(X)=E(X^2)=2\int_0^1 x^2(1-x)dx=2(\frac{1}{3}-\frac{1}{4})=\frac{1}{6}$.

5 $P(X<980)=P(\frac{X-1000}{s}<\frac{980-1000}{s})\leq 0.02$. The left hand side is standard normal, so we have $\Phi\left(\frac{980-1000}{s}\right)\leq 0.02$. It means that $\frac{980-1000}{s}\leq -2.0535$, so we get $s\leq 9.74$.

6 $(3X-cY,2X+Y)=6Var(X)-cVar(Y)$ as the other terms are equal to 0. $Var(X)=Var(Y)=1$ implies that $c=6$ is the solution.

Example Midterm

6 What is shown by the following R-code? What is given by p? What is the theoretical value of p?

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm
temp = np.random.normal(loc=1, scale=2, size=1000)
plt.figure(figsize=(10, 6))
plt.hist(temp, bins=30, color="orange", density=True, edgecolor="black")
xvalues = np.linspace(-6, 8, 100)
yvalues = norm.pdf(xvalues, loc=1, scale=2)
plt.plot(xvalues, yvalues, color="blue", linewidth=3)
plt.title("Normal distribution")
plt.xlabel("values")
plt.ylabel("frequencies (density)")
plt.grid(True)
plt.show()
p = np.sum((-1 < temp) & (temp < 3)) / len(temp)
print(f"The calculated proportion p is:  {p}")
```

Example Midterm

6 irst we generate 1000 random numbers with normal distribution (with mean=1 and sd=2). Next we visualise them by a density-histogram (in orange color) and add a curve of the theoretical density function of the normal distribution. Finally, p gives the number of observations that were in the interval $(-1; 3)$.
The theoretical value of p is
$P(X < 3) - P(X > -1) = P(\frac{X-1}{2} < \frac{3-1}{2}) - P(\frac{X-1}{2} > \frac{-1-1}{2}) = \Phi(1) - \Phi(-1) = \Phi(1) - (1 - \Phi(1)) = 2\Phi(1) - 1 = 2 \cdot 0.8413 - 1 = 0.6826$